OPEN PLATFORM FIRMWARE (OPF)



The Next Evolution of Intel's FSP: Aligning to openSFI and UPCI

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Acknowledgement

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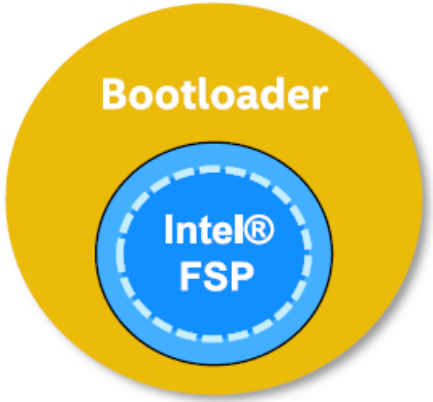
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Agenda

- Intel® FSP Overview
- Data Center Intel® FSP Enhancement
- openSFI and UPCI Alignment
- Call To Action

Intel® FSP Overview

- Intel® FSP is binary encapsulation of Intel's silicon initialization code
- Intel® FSP continues to be primary vehicle to enable Intel silicon - currently at V2.5 with support for 64bit execution environment and SMM.
  https://www.intel.com/content/www/us/en/content-details/824483/content-details.html
- Intel® FSP supports various bootloaders (UEFI, coreboot, Slim Bootloader) through API and Dispatch modes.



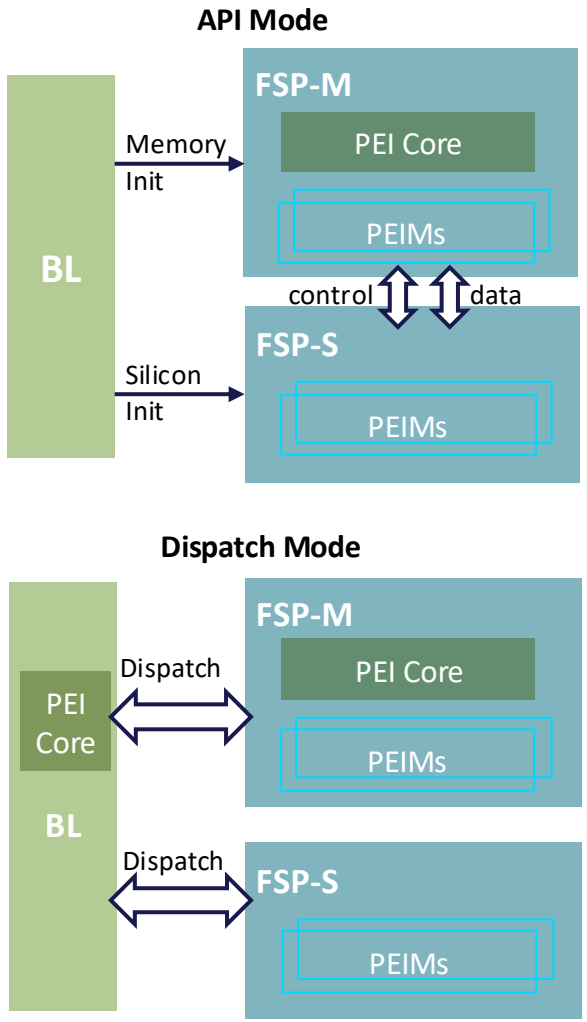
Data Center Intel® FSP Enhancement

- Simplify FSP-M design by making it independent of EDKII framework.
- FSP-S independent of FSP-M.
- Support API mode by default and launch dispatch mode environment opportunistically.
- Active engagement in OpenSFI definition.

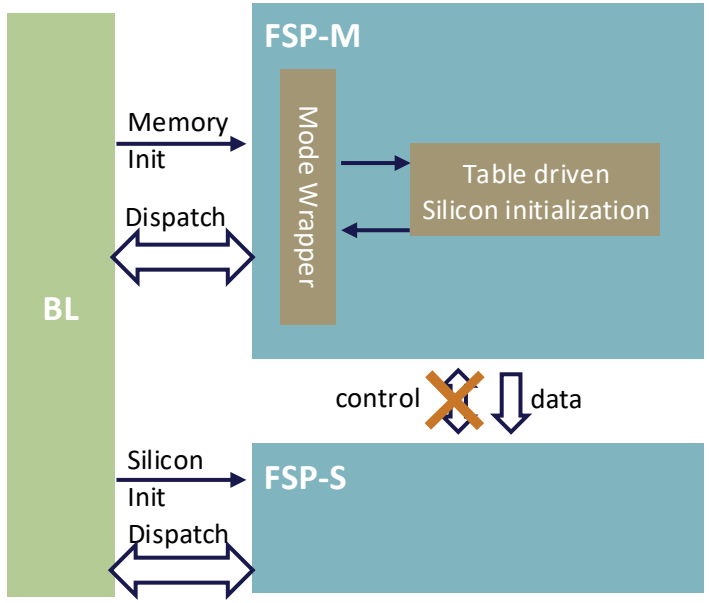
❖No Intel® FSP specification changes – no impact to bootloader code

Data Center Intel® FSP enhancement - High Level

Intel® Xeon® 6



Next Intel® Xeon® generation

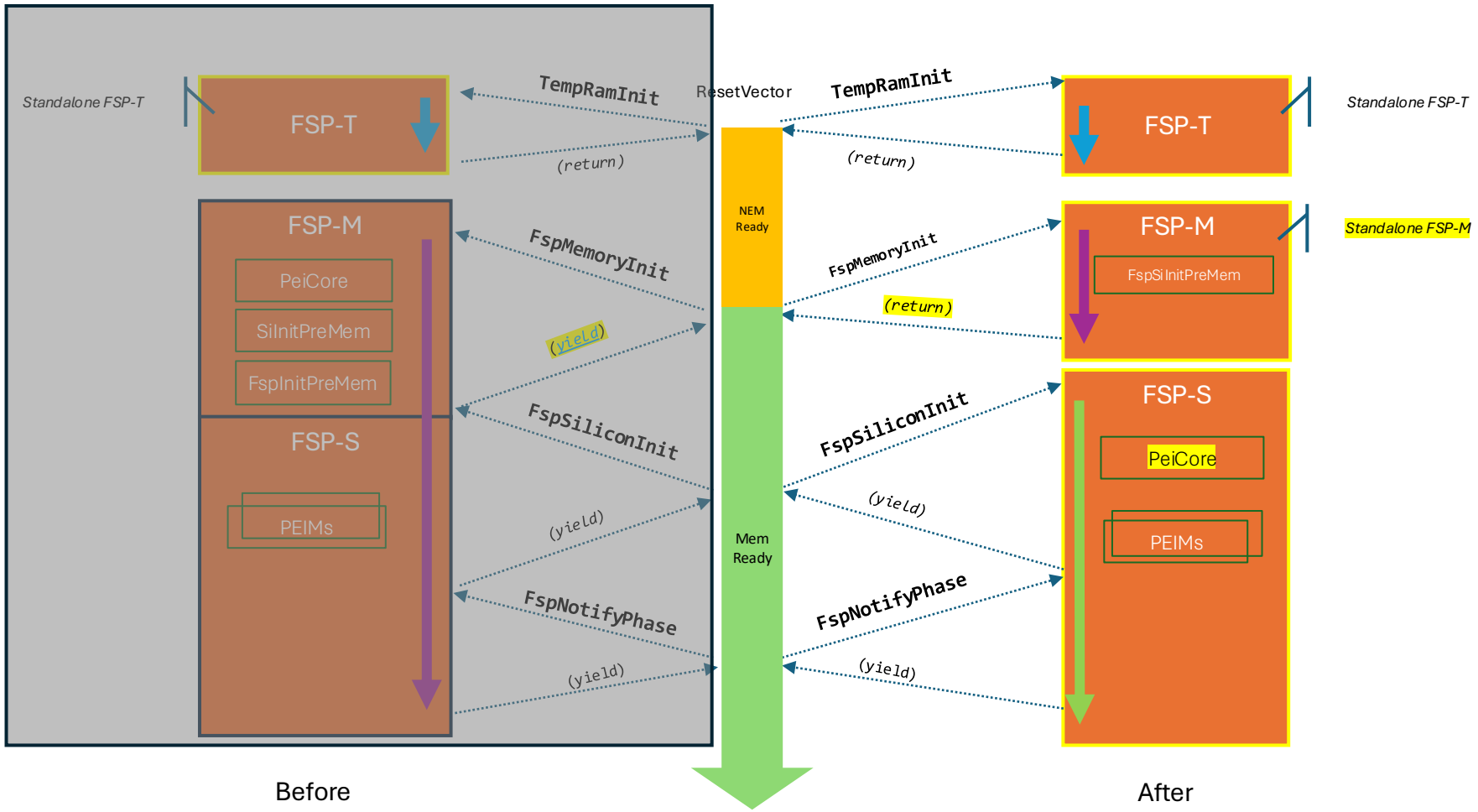


- Converged API and Dispatch mode flow
- Simplify FSP-M to link **silicon code libraries** into single module **(no PEI dependency)**
- **Separate FSP-M and FSP-S execution**

OpenSFI - Stateless execution model

- Avoid persistent state between calls.
- Initialization stages are “stateless” to host FW.
- The state is maintained by HostFW managed resources, and the OpenSFI Si code does not have other resources to save state info across calls.

OpenSFI alignment - Stateless execution model



Intel ® FSP fixes stateful to align to the principle of OpenSFI – Statelessness

- Decouple FSP-M and -S in FSP Entry Code
- Move PEI Core from FSP-M to FSP–S
- FSP-M runs in temporary RAM; FSP-S runs in physical memory

Call to Action

1. Prepare your OPF solutions for Intel® server FSP
   - Intel® FSP updates currently planned for a future Intel® Xeon® platform
2. Check out the Intel® server FSP update materials
   - Contact your Intel support for more details.
3. Collaborate with us continue to evolve Intel FSP opensource partnership
   - TianoCore/EDKII
   - coreboot (Intel coreboot)
   - Slim Bootloader (Slim Bootloader Project)

Thank You!

